



DoA Estimation for UCAs in Non-minimum Phase Channels

B.Tech Project

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Introduction

- ❑ A “**non-minimum phase**” channel is when non-LOS signal components arrive at Rx with higher power. This can be encountered in the form of **IRS/RIS**-aided spoofing in electronic warfare.
- ❑ Multipath and IRS/RIS-based interference create coherent sources, leading to **rank-deficiency** of the data covariance matrix in **subspace based methods**.
- ❑ Traditional model order selection (AIC/MDL) [1] criteria fail in this scenario.

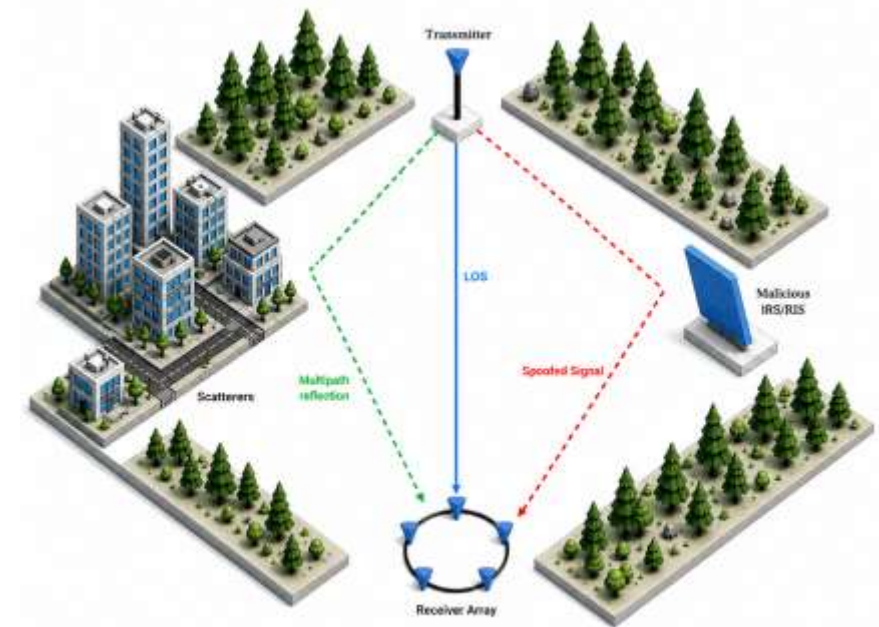


Fig. 1: Illustration of stationary jammer scenario

[1] M. Wax and T. Kailath, “Detection of signals by information theoretic criteria,” IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. 33, no. 2, pp. 387–392, 1985

Failure of AIC and MDL

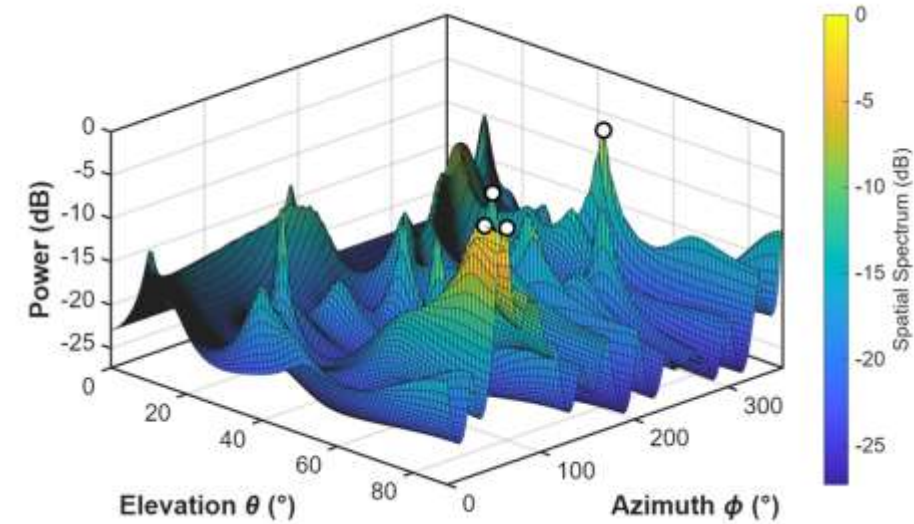


Fig. 2: Spatial Spectrum using AIC estimate

Ground Truth Signal Scenario:

Source	Φ (Az)	Θ (El)	Power
S1 (LOS)	55	90	-3 dB
S2	65	90	0 dB
S3	85	90	-6 dB
S4	100	90	-12 dB

Case	Model Order Selection Criterion	Estimated peaks (φ, θ)	Power (dB)
Unsmoothed	Manual	(65, 82)	0
		(273, 29)	-2.12
		(333, 89)	-2.41
Smoothed	AIC	(65, 90)	0
		(57, 85)	-0.15
		(101, 75)	-4.19
	MDL	(65, 90)	0
		(55, 89)	-0.08
		(101, 89)	-4.58
	Our Method	(55, 90)	0
		(62, 89)	-0.71
		(269, 26)	-1.98

Table I: Comparison of Dominant Peak Estimates Under Different Model-Order Selection Methods

Parameters:

Carrier: $f = 702$, Geometry: $N = 5$ elements, $r = 0.5095$ m, SNR: 9 dB, Snapshots: $T = 128$

The Beamspace-MUSIC Algorithm



□ Beamformer weighting scheme [2]:

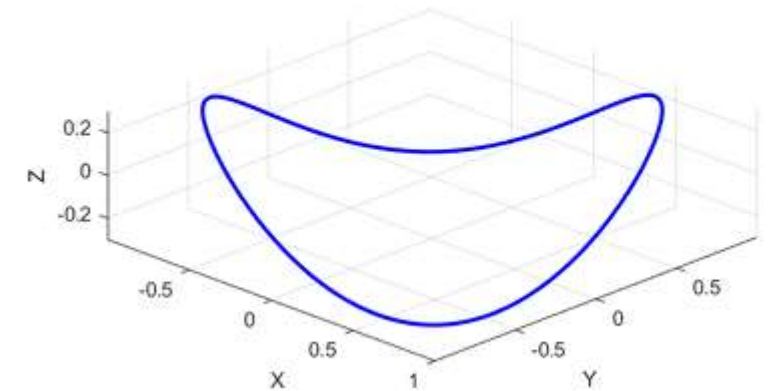
$$w(\gamma_n) = e^{jm\gamma_n}, \quad \text{where } \gamma_n = \frac{2\pi(n-1)}{N}$$

$$y_m(t) = \sum_{n=0}^{N-1} x_n(t) \cdot e^{jm\gamma_n}$$

□ Far-field array pattern for mode 'm' ($N > 2M$) becomes $A_m(\phi) = j^m \cdot J_m(\zeta) \cdot e^{jm\phi}$

□ After the **beamspace transformation**, the UCA manifold is mapped onto an M' -dimensional **virtual ULA**. The resulting virtual ULA is partitioned into $S = M' - M'' + 1$ overlapping M'' -dimensional subarrays for mode-space **spatial smoothing** [3].

Mode 2: Sine Wave Wrapped Around a Circle (2 Cycles)



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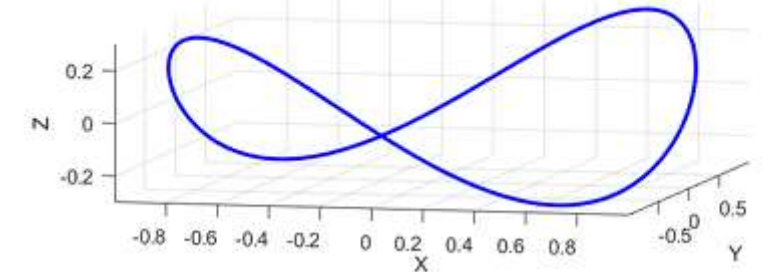


Fig. 3: An illustration of the “spatial harmonic” modes.

[2] C. Mathews and M. Zoltowski, “Eigenstructure techniques for 2-d angle estimation with uniform circular arrays,” IEEE Transactions on Signal Processing, vol. 42, no. 9, pp. 2395–2407, 1994.

[3] R. Eiges and H. Griffiths, “Mode-space spatial spectral estimation for circular arrays,” IEE Proceedings - Radar, Sonar and Navigation, vol. 141, pp. 300–306, 1994

Dynamic Thresholding

$$C_{ij} = \begin{cases} 1, & i = j \\ \rho_f(\theta_i, \theta_j), & i \neq j \end{cases} \quad \left(|\rho_f| = \left| \rho \frac{\sin(S(\theta_i - \theta_j)/2) \cos \psi}{S \sin((\theta_i - \theta_j)/2)} \right| \right)$$

$$\overline{R}_s = P^{1/2} C P^{1/2}$$

$$\text{Tr}(\overline{R}_s^{-1}) = \sum_{i=1}^D \frac{1}{\lambda_i} \approx \frac{1}{\lambda_D(\overline{R}_s)}$$

$$\lambda_D(\overline{R}_s) \approx \left\{ \text{Tr}(P^{-1/2} C^{-1} P^{-1/2}) \right\}^{-1} = \left\{ \sum_{i=1}^D \frac{[C^{-1}]_{ii}}{P_i} \right\}^{-1}$$

$$\lambda_D(R_x) = \lambda_D(A_s \overline{R}_s A_s^H) + \sigma_n^2$$

$$\lambda_D(R_x) \approx (M' - S + 1) \left\{ \sum_{i=1}^D \frac{P_i}{[C^{-1}]_{ii}} \right\} + \sigma_n^2$$

$$(M' - S + 1) \left\{ \sum_{i=1}^D \frac{P_i}{[C^{-1}]_{ii}} \right\} \geq (K - 1) \sigma_n^2$$

C_{ij} : (DxD) post-smoothing Source Correlation Matrix

$|\rho_f|$: effective correlation coefficient

Dynamic noise variance estimate:

$$\hat{\sigma}_n^2 = \frac{1}{p} \sum_{i=M''-p+1}^{M''} \lambda_i$$

Signal Sub-space Boundary definition:

$$\gamma = K \cdot \hat{\sigma}_n^2$$

- Eigenvalues satisfying $\lambda_i > \gamma$ are classified as belonging to the restored signal subspace.

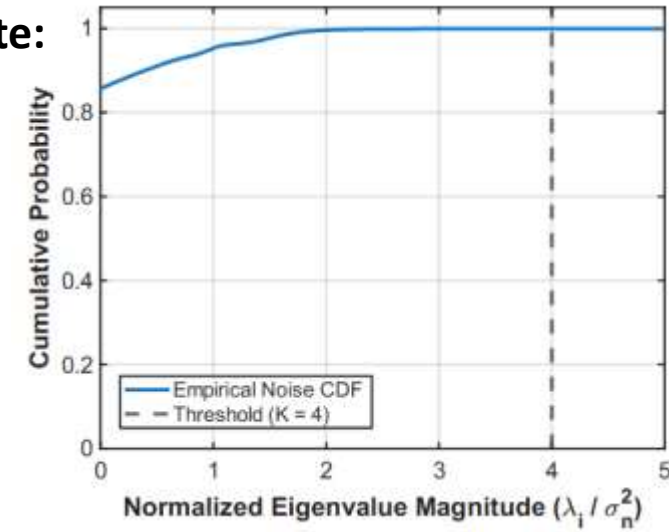


Fig. 4: Empirical CDF of normalized noise eigenvalues after phase-mode smoothing and forward-backward averaging

Results: Smoothed vs Un-smoothed

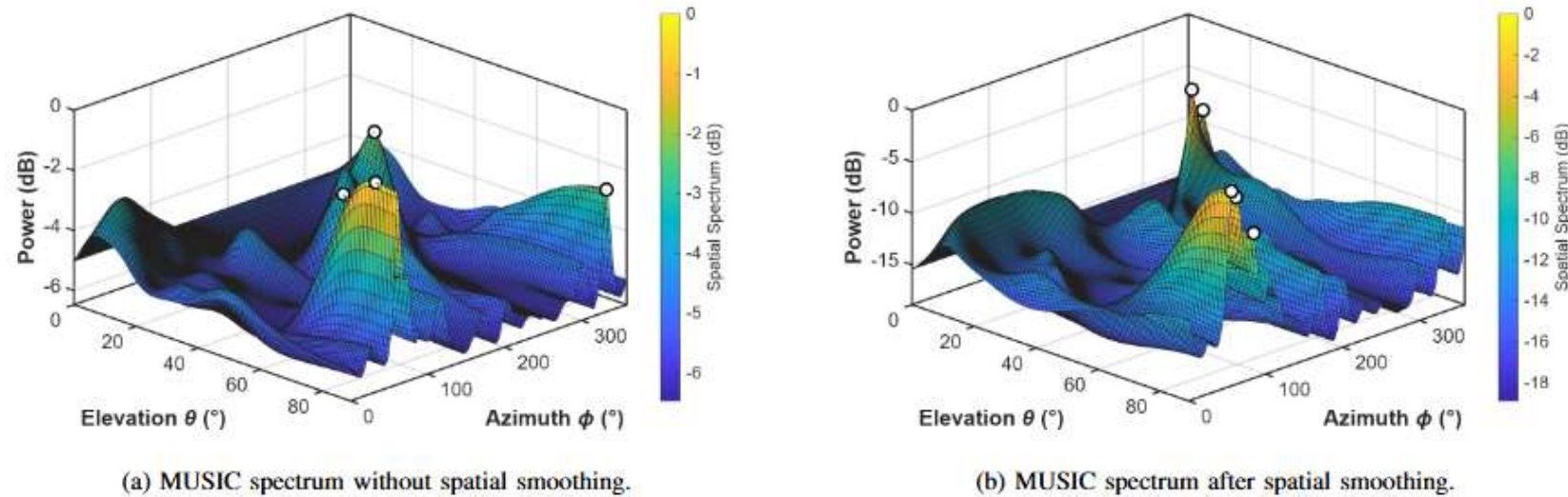


Fig. 5: Comparison of MUSIC pseudo-spectra in the presence of coherent multipath.

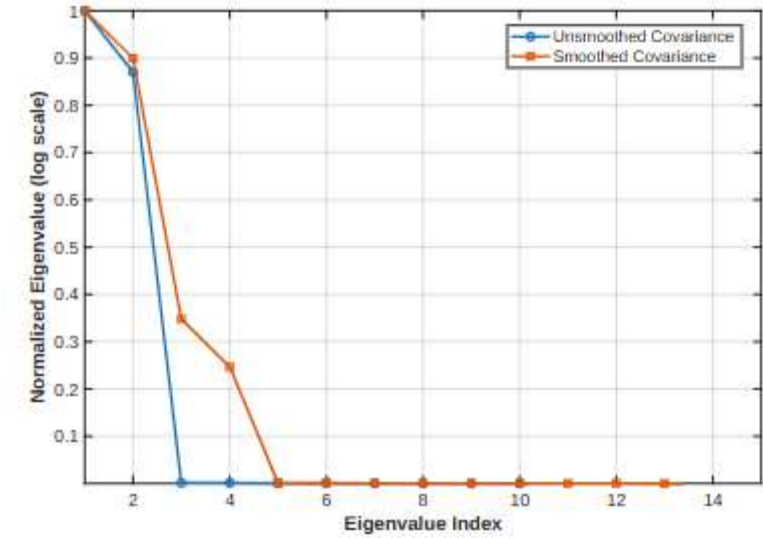


Fig. 6: Eigenvalue distribution before and after spatial smoothing

- ❑ As summarized in table I and Fig. 5, the smoothed variant with dynamic thresholding successfully isolates LOS DoA as the dominant peak.
- ❑ After smoothing, four dominant eigenvalues became clearly separable enabling proper subspace partitioning using the proposed thresholding scheme (Fig. 6).

RMSE vs SNR and Jammer Strength

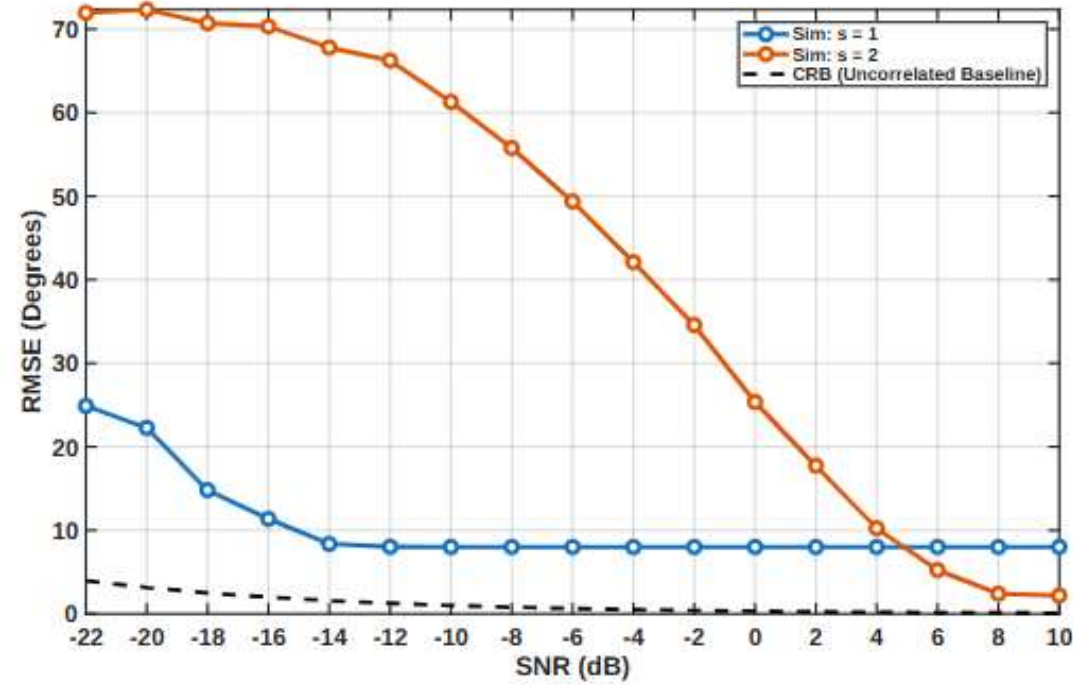


Fig. 7: RMSE comparison between standard UCA-RB-MUSIC ($s = 1$) and the proposed mode-space smoothed variant ($s = 2$) against the theoretical CRB.

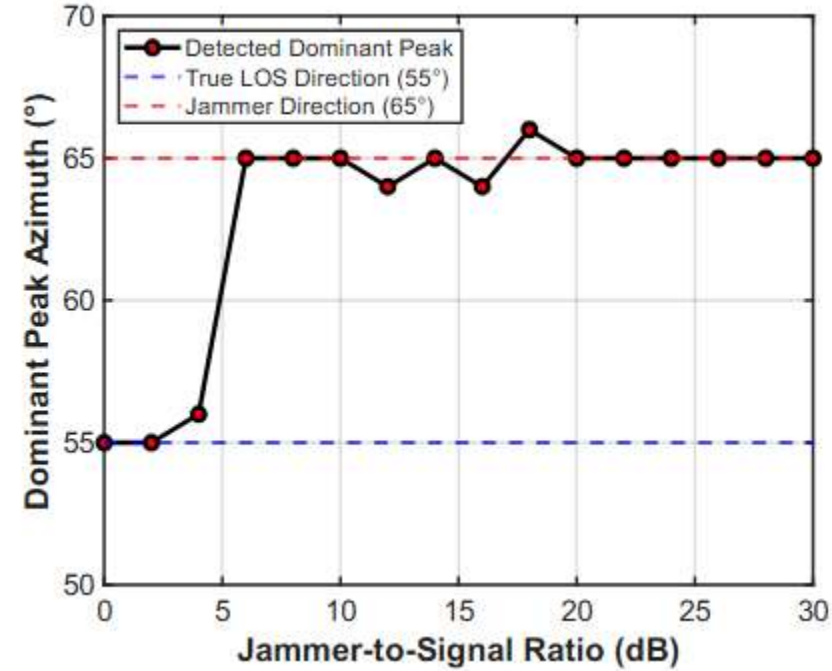


Fig. 8: Dominant detected azimuth as a function of jammer-to-signal ratio(JSR).

- Fig. 7 shows the performance of the algorithm over a range of SNRs. Each point represents an average of 400 Monte-Carlo trials. Beyond 5 dB operational SNR, our method approaches the CRB.
- Fig. 8 quantifies the operational limits of the proposed algorithm, demonstrating an observed tolerance of up to +4 dB.

Channel Variations



Scenario	Channel Condition	Delay	Peak (φ, θ)
Open Terrain Combat	Specular-dominant	Short	(55, 90)
Open Terrain Combat	Specular-dominant	Medium	(55, 90)
Open Terrain Combat	Specular-dominant	Long	(65, 90)
Urban Combat	Fully scattered	Short	(55, 90)
Urban Combat	Fully scattered	Medium	(65, 90)
Urban Combat	Fully scattered	Long	(65, 90)

Table III: First Peak Estimates across Open Terrain Combat Zone and Urban Combat Zone Scenarios

- ❑ Table II summarizes localization across both channel conditions. Longer delays cause spatial decorrelation which in turn degrades the algorithm performance.

Real-data validation

Freq (MHz)	Standard (Az, El, P)	With FBA (Az, El, P)	Mode-Space Smoothing (Az, El, P)
252	322°, 90°, -5.7	320°, 90°, 0.0	340°, 73°, 0.0
412	152°, 19°, -2.6	320°, 90°, 0.0	320°, 86°, 0.0
622	52°, 30°, -4.7	147°, 36°, 0.0	317°, 77°, 0.0
647	102°, 17°, 17.4	149°, 33°, 0.0	316°, 72°, 0.0
702	150°, 23°, -2.3	319°, 90°, 0.0	319°, 89°, 0.0
712	148°, 23°, 0.4	322°, 90°, 0.0	322°, 89°, 0.0
707	-	149°, 23°, 3.0	320°, 89°, 0.0
812	161°, 14°, 2.3	163°, 12°, 0.0	320°, 89°, 0.0

Table III: 5-element UCA DoA detection performance using field data.



Fig. 9: The 5-element UCA field deployment utilized for real-time data acquisition

❑ The source during the field test was located at 320°. Table III summarizes the performance and Fig. 9 shows the 5-element UCA provided by Data Patterns.*

* Ownership of data and proprietary IP lies with TelWise, IIT Madras and Data Patterns, Chennai.

Conclusion



- A narrow operating regime allows for a potential solution to a rather non-trivial problem in electronic warfare.
- **Future work:** Further real-data validation with a nine-element Uniform Circular Array.



Thank You!